

RESEARCH INTERESTS

I study autonomous agents and multi-agent interaction, with a focus on how representations and constraints shape agent behavior. I am particularly interested in connecting agent architectures with human-AI interaction and world models, and in building agent systems that are more interpretable, controllable, and trustworthy.

RESEARCH EXPERIENCE

Human-Centered Emergency Preparedness Research, Columbia University
Research Assistant with Prof. Harry West, Preparedness & Recovery Institute Summer 2026-Present

- Develop product concepts, interaction frameworks, and iterative prototypes from research findings, focusing on rapid communication, community activation, and emergency decision support.
- Conduct cross-case synthesis across interviews, workshops, and stakeholder engagements with city agencies, community organizations, and local leaders, identifying recurring breakdowns in rapid consultation, information handoffs, implementation, and feedback under emergency conditions.
- Conduct interviews, workshops, and prototype testing with public-health, emergency-preparedness, and community stakeholders, including participants from NYC DOHMH, CDC, and other government and community organizations, examining how community and frontline knowledge can better inform time-sensitive institutional decision-making.

Undergraduate Researcher, Software Systems & Security Columbia University
Advisor: Prof. Zhuo Zhang Spring 2026

- Investigated the architecture, execution behavior, and security properties of tool-using AI agents, using Nanobot and Claude Code as case studies. Developed a research direction around replayable execution traces for agent analysis, debugging, and evaluation.

Digital Storytelling Lab (DSL), Columbia University 2025-Present

- Participated in project-based research on interactive systems and computational storytelling, with an emphasis on iterative prototyping and analysis of agent behavior, user interaction, and narrative dynamics.
- Selected demos: [\[YouTube link\]](#)

Columbia DevFest Hackathon Columbia University
Participant Nov. 2025

- Participated in rapid prototyping of interactive AI systems; responsibilities included prompt engineering, narrative design, and interface logic.

NSF(National Science Foundation) I-Corps Princeton University
Propelus Certificate of Completion Spring 2024

- Completed National Science Foundation (NSF) Innovation Corps (I-Corps™) Northeast Hub-funded training in research commercialization, customer discovery, and early-stage product validation.
- Received micro-funding to identify initial customers and define a prototype concept, collaborating with entrepreneurs and mentors across NSF's innovation network.
- Designed an AI system informed by Buddhist contemplative principles to support reflective user interaction. Secured NSF I-Corps funding and conducted over 100 user interviews to validate needs among monasteries, practitioners, and scholars.

SELECTED RESEARCH PROJECTS

Interactive Asylum Case-Review System Columbia University
Migration between France and North Africa, Professor Madeleine Dobie Summer 2026

- Designed an interactive asylum case-review system that placed users in the role of an adjudicator reviewing a fictional case file. The interface explicitly warned that the evidence was incomplete but still required a decision, with no option to abstain—turning evidentiary uncertainty and institutional obligation into an interaction rather than simply describing them. [\[Project link\]](#)

Emotion Landscape — Interactive Storytelling System

Columbia DSL (Digital Storytelling Lab)

New York City, USA
2025-Present

- Investigated how structured prompting strategies affect emergent behavior in interactive, agent-driven narrative environments. Designed modular interaction logic to study coordination between narrative agents and user actions.
- Built an assets pipeline and explored affective design patterns for interactive spaces.

Prompt Markdown+

Independent Research

2025-Present

- Designed a structured prompt language to improve controllability and interpretability, and to study how abstraction layers affect agent failure modes and user mental models in human–AI interaction.
- Built an early prototype and evaluation plan to measure agent consistency under constrained prompt schemas (preparing user study).

Interactive Engine (In.js)

Independent Research

2025-Present

- Built a modular JavaScript-based interactive engine to experiment with agent-driven narrative logic.
- Exploring how architectural choices affect scalability and agent coordination in interactive systems.

EDUCATION

Columbia University

B.A. in Computer Science & Cognitive Science

New York City, USA

- GPA 4.0 / 4.0
- 2025-Present, Expected 2029

Chinese Academy of Social Sciences (CASS)

M.Phil., Humanities (textual and visual analysis; narrative structures)

Beijing, China

- Overall GPA: 3.70/4.0

Beijing University of Technology (BJUT)

Bachelor of Engineering, Civil Engineering

Beijing, China

ACADEMIC HONORS & AWARDS

Charlotte W. Newcombe Foundation Endowed Scholarship, Columbia University

2026-27

- Selected for a named endowed scholarship reserved for the School’s most talented current students.

Millennium Fellow, Millennium Campus Network & United Nations Academic Impact

2026-27

- Selected for the U.S. & MENA Region Virtual Exchange Program.

Global Learning Scholarship, Columbia University

Dean’s List, Columbia University School of General Studies

Exceptional Master’s Thesis, Chinese Academy of Social Sciences

Summer 2026
All eligible terms
2011

SELECTED PRODUCT & TECHNOLOGY EXPERIENCE

- Developed products and prototypes across fintech, digital publishing, blockchain infrastructure, and AI, translating complex technical ideas into clear, testable user experiences.
- Worked across web, iOS, and Android, combining product strategy, interface design, technical feasibility, and cross-cultural project coordination.

TECHNICAL SKILLS

- Programming: Python, Java, C/C++, JavaScript
- Systems/Dev: Linux, Git, CLI
- AI/Agents: Multi-agent Prompting, HCI Prototyping
- Prototyping & Media: Interactive demos, Video-based documentation, Godot